C

ORDINARY DIFFERENTIAL EQUATIONS • UNIT 8.01

Power Series Solutions & Method of Frobenius

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

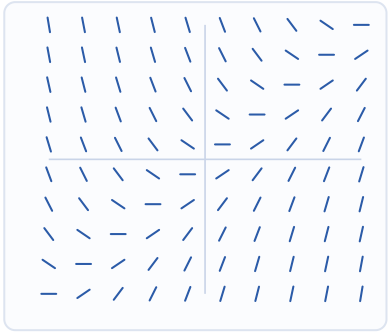
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 B	Q3 B	Q4 C	Q5 A
Q6 A	Q7 D	Q8 A	Q9 B	Q10 A
Q11 D	Q12 B	Q13 C	Q14 B	Q15 D
Q16 A	Q17 D	Q18 A	Q19 C	Q20 B
Q21 C	Q22 C	Q23 A	Q24 A	Q25 D

Power Series Solutions & Method of Frobenius Practice Exercises

1 MCQ

For $(x^2 - x)y'' + y' - y = 0$, the point $x = 0$ is:



A ordinary point

B a point where no solution exists

C **singular point** ✓

D a point where every solution vanishes

WORKED SOLUTION & VERIFICATION

Dividing by $x^2 - x$ gives $y'' + P(x)y' + Q(x)y = 0$. The leading coefficient vanishes at $x = 0$, so P, Q are not analytic there: it is a singular point.

2 MCQ

Near $x_0 = 0$, the power series solutions of $(x + 4)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

A 5

B 4 ✓

C ∞

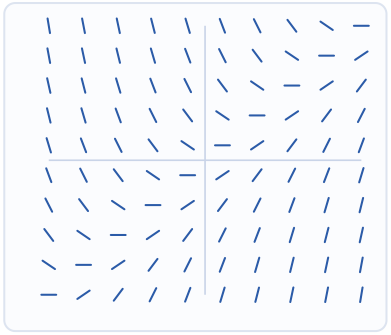
D 1

WORKED SOLUTION & VERIFICATION

The singular points are the zeros of $x + 4$: -4 (complex ones count). The nearest is at distance 4 from 0.

3 MCQ

For $(x^2-9)y''+y'-y=0$, the point $x=-1$ is:



- **A** singular point
- **B** **ordinary point** ✓
- **C** a point where no solution exists
- **D** a point where every solution vanishes

WORKED SOLUTION & VERIFICATION

Dividing by x^2-9 gives $y''+P(x)y'+Q(x)y=0$. P and Q are analytic at -1 iff the leading coefficient is nonzero there; $x^2-9=-8$, so it is an ordinary point.

4 MCQ

Near $x_0=0$, the power series solutions of $(x^2-4x+3)y''+xy'+y=0$ are guaranteed to converge at least for $|x-x_0|<R$ with $R=$

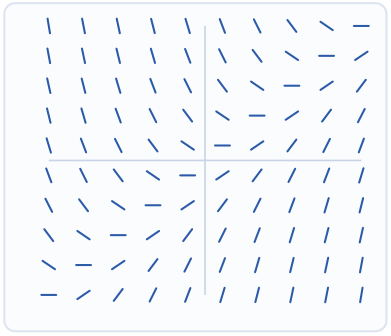
- **A** 3
- **B** ∞
- **C** 1 ✓
- **D** 2

WORKED SOLUTION & VERIFICATION

The singular points are the zeros of x^2-4x+3: $1, 3$ (complex ones count). The nearest is at distance 1 from 0.

5 MCQ

For $(x^2+x)y''+y'-y=0$, the point $x=-1$ is:



- **A** **singular point** ✓
- **B** ordinary point
- **C** a point where no solution exists
- **D** a point where every solution vanishes

WORKED SOLUTION & VERIFICATION

Dividing by x^2+x gives $y''+P(x)y'+Q(x)y=0$. The leading coefficient vanishes at $x=-1$, so P, Q are not analytic there: it is a singular point.

6 MCQ

Near $x_0 = 0$, the power series solutions of $(x^2+4)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

- **A** 2 ✓
- **B** ∞
- **C** 1
- **D** 4

WORKED SOLUTION & VERIFICATION

The singular points are the zeros of x^2+4: $-2i, 2i$ (complex ones count). The nearest is at distance 2 from 0.

7 MCQ

For $(x^2 - 4x)y'' + y' - y = 0$, the point $x = 0$ is:

- **A** ordinary point
- **B** a point where no solution exists
- **C** a point where every solution vanishes
- **D** **singular point** ✓

WORKED SOLUTION & VERIFICATION

Dividing by $x^2 - 4x$ gives $y'' + P(x)y' + Q(x)y = 0$. The leading coefficient vanishes at $x = 0$, so P, Q are not analytic there: it is a singular point.

8 MCQ

Near $x_0 = 0$, the power series solutions of $(x^2 + x - 6)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

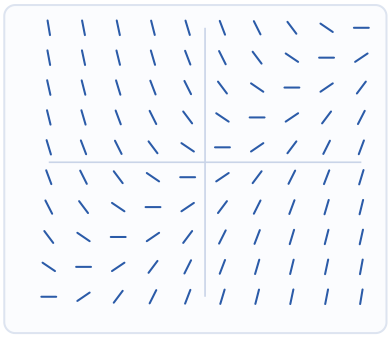
- **A** 2 ✓
- **B** 3
- **C** ∞
- **D** 1

WORKED SOLUTION & VERIFICATION

The singular points are the zeros of $x^2 + x - 6$: $-3, 2$ (complex ones count). The nearest is at distance 2 from 0.

9 MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.



A $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

B $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$ ✓

C $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

D $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

WORKED SOLUTION & VERIFICATION

We are given the differential equation $y'' - xy = 0$ and assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$. \newline Differentiating twice with respect to x, we get: $y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$ and $y''(x) = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$. \newline Substituting these into the differential equation: $\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n = 0$. \newline We can rewrite the second term as $\sum_{n=0}^{\infty} c_n x^{n+1}$. To combine the series, we need the powers of x to be the same. Let $k = n-2$ in the first series, so $n = k+2$. When $n = 2, k = 0$. The first series becomes $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$. \newline Let $k = n+1$ in the second series, so $n = k-1$. When $n = 0, k = 1$. The second series becomes $\sum_{k=1}^{\infty} c_{k-1} x^k$. \newline So the equation is $\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline We can separate the $k = 0$ term from the first sum: $2(1)c_2 + \sum_{k=1}^{\infty} (k+2)(k+1) c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k = 0$. \newline For this equation to hold for all x, the coefficients of each power of x must be zero. \newline For $k = 0$: $2c_2 = 0$ implies $c_2 = 0$. \newline For $k \geq 1$: $(k+2)(k+1)c_{k+2} - c_{k-1} = 0$. \newline Thus, the recurrence relation is $c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}$ for $k \geq 1$. Replacing k with n, we get $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$.

10 MCQ

The Laplace transform of $h(u) = 190e^{20u}$ is $k(s) = \frac{190}{s-20}$ for $s > 20$. Determine the evaluated transform $k(21)$.

A 190 ✓

B 209

C 170

D 193

WORKED SOLUTION & VERIFICATION

Substituting $s = 21$ gives $\frac{190}{21-20} = 190$.

11 MCQ

Consider the second-order linear homogeneous differential equation: $x^2y'' + xy' + (x^2 - \nu^2)y = 0$, where ν is a constant. This is Bessel's equation of order ν. Assume $\nu = 1/2$ and that $x > 0$.

Part 1: Ordinary Point Analysis

$x = 0$ is a regular singular point. Determine the indicial equation for this differential equation and find its roots.

A Indicial equation: $r^2 + \frac{1}{4} = 0$; roots: $r = \pm\frac{i}{2}$

B Indicial equation: $r^2 - r + \frac{1}{4} = 0$; roots: $r = \frac{1}{2}, \frac{1}{2}$

C Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{4}$

D **Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{2}$** ✓

WORKED SOLUTION & VERIFICATION

The given differential equation is $x^2y'' + xy' + (x^2 - \nu^2)y = 0$. We can rewrite this in the standard form $y'' + P(x)y' + Q(x)y = 0$ by dividing by x^2: $y'' + \frac{1}{x}y' + \frac{x^2-\nu^2}{x^2}y = 0$. Thus, $P(x) = \frac{1}{x}$ and $Q(x) = \frac{x^2-\nu^2}{x^2} = 1 - \frac{\nu^2}{x^2}$. $x = 0$ is a singular point. To determine if it's a regular singular point, we examine $xP(x)$ and $x^2Q(x)$. $xP(x) = x \cdot \frac{1}{x} = 1$, which is analytic at $x = 0$. $x^2Q(x) = x^2 \cdot \frac{x^2-\nu^2}{x^2} = x^2 - \nu^2$, which is also analytic at $x = 0$. Therefore, $x = 0$ is a regular singular point. The method of Frobenius assumes a solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^{n+r}$, where $c_0 \neq 0$. Substituting this into the differential equation, we get the indicial equation, which is a quadratic equation in r. For Bessel's equation, the indicial equation is given by $r(r-1) + p_0 r + q_0 = 0$, where $p_0 = \lim_{x\to 0} xP(x)$ and $q_0 = \lim_{x\to 0} x^2Q(x)$. In our case, $p_0 = 1$ and $q_0 = -\nu^2$. So the indicial equation is $r(r-1) + 1 \cdot r - \nu^2 = 0$, which simplifies to $r^2 - r + r - \nu^2 = 0$, or $r^2 - \nu^2 = 0$. Given $\nu = 1/2$, the indicial equation becomes $r^2 - (1/2)^2 = 0$, which is $r^2 - \frac{1}{4} = 0$. The roots of this equation are $r = \pm\sqrt{\frac{1}{4}} = \pm\frac{1}{2}$.

12 MCQ

Consider the differential equation $y'' - xy = 0$. We are looking for a power series solution of the form $y(x) = \sum_{n=0}^{\infty} a_n x^n$ around the ordinary point $x = 0$. Find the coefficient a_2 in terms of a_0 and a_1.

A $a_2 = a_0 + a_1$

B $a_2 = 0$ ✓

C $a_2 = a_0$

D $a_2 = a_1$

WORKED SOLUTION & VERIFICATION

We assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} a_n x^n$. Then $y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ and $y''(x) = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$.

Substituting these into the differential equation $y'' - xy = 0$, we get:

$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - x\sum_{n=0}^{\infty} a_n x^n = 0$

$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=0}^{\infty} a_n x^{n+1} = 0.$

To compare coefficients, we need the powers of x to be the same. Let $k = n - 2$ in the first sum, so $n = k + 2$. When $n = 2$, $k = 0$. Let $k = n + 1$ in the second sum, so $n = k - 1$. When $n = 0$, $k = 1$.

$\sum_{k=0}^{\infty}(k+2)(k+1)a_{k+2}x^k - \sum_{k=1}^{\infty} a_{k-1}x^k = 0.$

We can split off the $k = 0$ term from the first sum:

$2(1)a_2x^0 + \sum_{k=1}^{\infty}(k+2)(k+1)a_{k+2}x^k - \sum_{k=1}^{\infty} a_{k-1}x^k = 0.$

$2a_2 + \sum_{k=1}^{\infty}[(k+2)(k+1)a_{k+2} - a_{k-1}]x^k = 0.$

For this equation to hold for all x, the coefficients of each power of x must be zero.

For the constant term (x^0), we have $2a_2 = 0$, which implies $a_2 = 0$.

For $k \geq 1$, we have the recurrence relation $(k+2)(k+1)a_{k+2} - a_{k-1} = 0$, or $a_{k+2} = \frac{a_{k-1}}{(k+2)(k+1)}$.

13 MCQ

The first-order differential equation $\frac{du}{ds} = 17s - 6u$ has its slope field plotted below. Determine the slope $\frac{du}{ds}$ at the point $K(14, 7)$.

A 189

B 213

C **196** ✓

D 199

WORKED SOLUTION & VERIFICATION

Evaluating at $(14, 7)$ gives $17(14) - 6(7) = 196$.

14 MCQ

For $(x^2 + x)y'' + y' - y = 0$, the point $x = -2$ is:

A singular point

B **ordinary point** ✓

C a point where no solution exists

D a point where every solution vanishes

WORKED SOLUTION & VERIFICATION

Dividing by $x^2 + x$ gives $y'' + P(x)y' + Q(x)y = 0$. P and Q are analytic at -2 iff the leading coefficient is nonzero there; $x^2 + x = 2$, so it is an ordinary point.

15 MCQ

Near $x_0 = 0$, the power series solutions of $(x^2 + 6)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

A ∞

B 1

C $2\sqrt{6}$

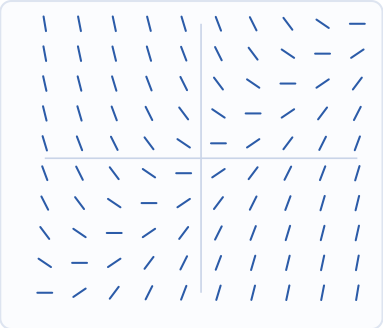
D $\sqrt{6}$ ✓

WORKED SOLUTION & VERIFICATION

The singular points are the zeros of $x^2 + 6$: $-\sqrt{6}i, \sqrt{6}i$ (complex ones count). The nearest is at distance $\sqrt{6}$ from 0 .

16 MCQ

For $(x^2 - 16)y'' + y' - y = 0$, the point $x = 4$ is:



- **A** **singular point** ✓
- **B** a point where no solution exists
- **C** a point where every solution vanishes
- **D** ordinary point

WORKED SOLUTION & VERIFICATION

Dividing by $x^2 - 16$ gives $y'' + P(x)y' + Q(x)y = 0$. The leading coefficient vanishes at $x = 4$, so P, Q are not analytic there: it is a singular point.

17 MCQ

For $(x^2 + 2x)y'' + y' - y = 0$, the point $x = 3$ is:

- **A** a point where no solution exists
- **B** a point where every solution vanishes
- **C** singular point
- **D** **ordinary point** ✓

WORKED SOLUTION & VERIFICATION

Dividing by $x^2 + 2x$ gives $y'' + P(x)y' + Q(x)y = 0$. P and Q are analytic at 3 iff the leading coefficient is nonzero there; $x^2 + 2x = 15$, so it is an ordinary point.

18 MCQ

For $(x^2 - x)y'' + y' - y = 0$, the point $x = 2$ is:

- **A** **ordinary point** ✓
- **B** singular point
- **C** a point where no solution exists
- **D** a point where every solution vanishes

WORKED SOLUTION & VERIFICATION

Dividing by $x^2 - x$ gives $y'' + P(x)y' + Q(x)y = 0$. P and Q are analytic at 2 iff the leading coefficient is nonzero there; $x^2 - x = 2$, so it is an ordinary point.

19 MCQ

Near $x_0 = 0$, the power series solutions of $(x^2+5)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$

A ∞

B 1

C $\sqrt{5}$ ✓

D $2\sqrt{5}$

WORKED SOLUTION & VERIFICATION

The singular points are the zeros of x^2+5: $-\sqrt{5}i, \sqrt{5}i$ (complex ones count). The nearest is at distance $\sqrt{5}$ from 0 .

20 MCQ

For $(x^2-16)y'' + y' - y = 0$, the point $x = -1$ is:

A singular point

B **ordinary point** ✓

C a point where no solution exists

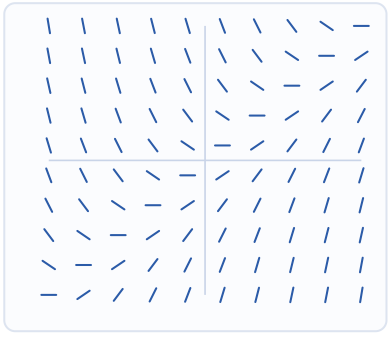
D a point where every solution vanishes

WORKED SOLUTION & VERIFICATION

Dividing by x^2-16 gives $y'' + P(x)y' + Q(x)y = 0$. P and Q are analytic at -1 iff the leading coefficient is nonzero there; $x^2 - 16 = -15$, so it is an ordinary point.

21 MCQ

Near $x_0 = 1$, the power series solutions of $(x-3)y'' + xy' + y = 0$ are guaranteed to converge at least for $|x - x_0| < R$ with $R =$



A 1

B 3

C 2 ✓

D ∞

WORKED SOLUTION & VERIFICATION

The singular points are the zeros of $x-3$: 3 (complex ones count). The nearest is at distance 2 from 1.

22 MCQ

For $(x^2-1)y''+y'-y=0$, the point $x=-1$ is:

- A ordinary point
- B a point where no solution exists
- C **singular point** ✓
- D a point where every solution vanishes

WORKED SOLUTION & VERIFICATION

Dividing by x^2-1 gives $y''+P(x)y'+Q(x)y=0$. The leading coefficient vanishes at $x=-1$, so P, Q are not analytic there: it is a singular point.

23 MCQ

For $(x^2-1)y''+y'-y=0$, the point $x=1$ is:

- A **singular point** ✓
- B ordinary point
- C a point where no solution exists
- D a point where every solution vanishes

WORKED SOLUTION & VERIFICATION

Dividing by x^2-1 gives $y''+P(x)y'+Q(x)y=0$. The leading coefficient vanishes at $x=1$, so P, Q are not analytic there: it is a singular point.

24 MCQ

Near $x_0=1$, the power series solutions of $(x+3)y''+xy'+y=0$ are guaranteed to converge at least for $|x-x_0|<R$ with $R=$

- A 4 ✓
- B 5
- C ∞
- D 1

WORKED SOLUTION & VERIFICATION

The singular points are the zeros of $x+3$: -3 (complex ones count). The nearest is at distance 4 from 1.

25 MCQ

Near $x_0=0$, the power series solutions of $(x+2)y''+xy'+y=0$ are guaranteed to converge at least for $|x-x_0|<R$ with $R=$

- A 3
- B ∞
- C 1
- D 2 ✓

WORKED SOLUTION & VERIFICATION

The singular points are the zeros of $x+2$: -2 (complex ones count). The nearest is at distance 2 from 0.